Multicollinearity

- **Multicollinearity** happens when two or more independent variables are highly related to each other.
- When variables are very similar, a regression model becomes confused about which variable is actually influencing the result.
- This can make model predictions unstable and difficult to interpret.

➢ To detect multicollinearity, we use **Variance Inflation Factor (VIF)**.

VIF Value	Interpretation
1	No multicollinearity
1-5	Moderate correlation but acceptable
5-10	High correlation
Above 10	Serious multicollinearity problem

How to handle multicollinearity?

1. **Remove one of the highly correlated variables.**

   Simple idea: If two friends say the same thing, you only need one of them.

   Example:

   hsc_p and etest_p highly correlated → remove one variable

2. **Combine similar variables into one variable.**

   Simple idea: Combine similar exam marks into one overall score.

   Example:

   Academic Score = average(ssc_p, hsc_p, degree_p)

3. **Use PCA (Principal Component Analysis)**

   PCA mixes many correlated variables together and creates new independent variables.

4. **Use Regularization Models**

   Some models automatically handle multicollinearity.

   These models reduce the impact of highly correlated variables.

   Example: Ridge Regression, Lasso Regression, Elastic Net

5. **Check the correlation matrix**

   Before modeling, check correlation between variables.

   If correlation is > 0.8 or < -0.8, variables are strongly related.